# EXP-10 IMAGE GENERATION USING GENERATIVE ADVERSARIAL NETWORK (GAN)

## AIM:

To generate synthetic images using a generative Adversarial network and evaluate their quality through visual inspection and suitable evaluation metrics

## SOFTWARE REQUIREMENTS:

* Python 3.x
* Pytorch
* Torchvision
* StyleGAN2-ADA
* CelebA Dataset
* Inception Score and FID evaluation tools.

## ALGORITHMS:

* Import the required libraries
* Set up the StyleGAN2-ADA repository
* Load a pretrained generator
* Generate random latent vectors
* Pass the latent vectors to the generator
* Generate Synthetic images
* Save the generated images
* Calculate the inception score or FID when sufficient real and generated images are available
* Compare the generated images with real images.

## PROGRAM:

```
import os
import torch
import matplotlib.plot as plt
```

## EXPECTED OUTPUT

Image generate successfully.

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

# Import StyleGAN
sys.path.append(os.path.abspath("stylegan2-ada-pytorch"))

import legacy
import dnnlib
from torch_utils.misc import save_image

with open(MODEL_PATH, "rb") as f:
    G = legacy.load_network_pkl(f)["G_ema"].to(device)

with torch.no_grad():
    img = G(z, label, truncation_psi=0.5, noise_mode="const")
    img = (img+1)/2

save_image(img, "generated_images/fake_0.png")
plt.imshow(img[0].permute(1, 2, 0).cpu().numpy())
plt.axis("off")
plt.title("Generated Image")
plt.show()

print("Image generated successfully")
```

## RESULT:

A pretrained GAN generator was used to generate Synthetic images from random latent vectors. The generated images can be evaluated visually and Compared with real images using metrics such as Inception Score and Fréchet Inception Distance (FID)